The finite-block cost of universality in entanglement extraction

Yongxian Zhang*

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Abstract

We study entanglement extraction by a global unitary on an existing many-copy input followed by deterministic local channels. The global preprocessing is an explicit entangling resource; no auxiliary system is supplied before it. We establish finite-dimensional spectral and subspace packing principles that match converses for completely PPT postprocessing. One state-independent protocol attains the state-aware capacity, exact fidelity exponent, strict-direct reliability function, and entropy-limited Gaussian threshold of every fixed input. A finite-block information-spectrum sandwich retains its dimension dependence and gives sufficient conditions for growing-dimensional Gaussian limits. We then determine the sharp minimax cost of an unknown support. For every fixed rank budget, a coherent common-support code matches a Haar-average converse, relatively and uniformly over all target dimensions. The support dimension and the reciprocal purity of the averaged input have the same leading asymptotic, yielding an explicit logarithmic loss in the extractable target size and a transition at the local-dimension cap. For balanced unknown pure inputs, the full-output fidelity has leading constant one. These results distinguish pointwise asymptotic universality from uniform finite-block performance. Standard compression, representation theory, and distillation results are explicitly credited.

1 Introduction and operational model

Universal quantum protocols operate without a classical description of their input. Universal compression, pure-state concentration, and mixed-state distillation establish several senses in which ignorance need not change an asymptotic rate [3, 5, 8, 9, 10, 11]. Equality of rates, however, does not determine a common protocol's finite-block performance. The same leading capacity can coexist with a polynomial fidelity loss, and a rank promise can incur a logarithmic target-size penalty invisible to the first-order rate.

We examine these distinctions in a two-stage extraction model. First apply one global unitary on the existing input space. Then apply deterministic local product channels, LOCC,

*School of Computer Science and Engineering, South China University of Technology, Guangzhou, China. Email: mxymmxym1@gmail.com. ORCID: 0009-0000-3864-3536.

or completely PPT channels. The global unitary is an entangling resource. This task is therefore distinct from ordinary LOCC distillation of the original fixed state. Its utility is to separate the spectral supply of purity from subsequent extraction and to ask what can be achieved by one preprocessing independent of the entire state.

The main method is a finite-dimensional comparison between spectral or common-subspace packing and the quadratic constraint on a completely PPT fidelity effect. It has three consequences. First, a variable-size Bell packing gives a one-shot spectral envelope, within logarithmic factors, for every bipartite spectrum. Second, a coherent simultaneous Schur packing gives one protocol for all states, with an explicit finite-block information-spectrum sandwich. Third, for a class of unknown inputs, a common-support code is compared with an adversarial averaged state. When the common-support dimension and the reciprocal purity of that average agree asymptotically, the entire minimax fidelity profile follows with leading constant one. Rank-bounded iid inputs provide a nontrivial instance of this last principle.

1.1 Model and quantifiers

Fix integers $2 \le m \le n$, set $D = mn$, and write

$$a = m^k, \qquad b = n^k, \qquad N = D^k$$

at copy number $k \ge 1$. All logarithms are natural. Let Φ_K denote the normalized rank-one maximally entangled projector on $\mathbb{C}^K \otimes \mathbb{C}^K$. We use the target overlap $\mathrm{Tr}(\Phi_K \sigma)$ as fidelity; no square root is taken.

A protocol comprises a unitary U on $\mathbb{C}^a \otimes \mathbb{C}^b$ and a trace-preserving channel Λ to $\mathbb{C}^K \otimes \mathbb{C}^K$. No ancilla, measurement, or discarded register is supplied before U. Local ancillas and discarding are permitted afterwards. The operation class LO consists of product channels. A channel is completely PPT, denoted cPPT, if both Λ and $\Gamma_{\mathrm{out}} \Lambda \Gamma_{\mathrm{in}}$ are completely positive and trace preserving; Γ transposes the second party. Thus LO $\subseteq$ LOCC $\subseteq$ cPPT. This definition is stronger than merely preserving the set of PPT states.

For a density matrix ρ on $\mathbb{C}^m \otimes \mathbb{C}^n$, define

$$f_k^{\mathcal{C}}(\rho, K) = \sup_{U, \Lambda \in \mathcal{C}} \mathrm{Tr}\Big[\Phi_K \Lambda(U \rho^{\otimes k} U^*)\Big]. \tag{1}$$

The optimizing protocol in this expression may depend on ρ. In contrast, our all-state universal family has quantifiers

$$\exists \{U_{k,K}, \mathcal{A}_{k,K}, \mathcal{B}_{k,K}\}_{k,K} \quad \forall \rho, \tag{2}$$

with dependence only on m, n, k, K. We write $F_k^{\mathrm{u}}(\rho, K)$ for its fidelity and $F_k^\star = f_k^{\mathrm{cPPT}}$. No spectrum, eigenbasis, rank, or entropy is an input to this family.

A different, explicitly promised task fixes a rank budget $1 \le s \le D$ and defines

$$\mathcal{G}_{\mathcal{C}}(k, K; s) = \sup_{U, \Lambda \in \mathcal{C}} \inf_{\mathrm{rank}\, \rho \le s} \mathrm{Tr}\Big[\Phi_K \Lambda(U \rho^{\otimes k} U^*)\Big]. \tag{3}$$

Here one protocol may depend on s, but it must serve every spectrum and every support orientation in that class. Pointwise universality in (2) and minimax universality in (3) are different assertions. We retain that distinction throughout.

1.2 Results and relation to prior work

Section 2 proves the actual finite-dimensional converse and packing principles. Section 3 gives a universal physical construction and a dimension-explicit finite-block sandwich. Section 4 derives the exact capacity and error laws. Section 5 proves the sharp rank-budget minimax profile, including the boundary at $n = ms$ and its logarithmic correction. Section 6 connects the capacity to APPT entropy optimization and specifies the scope of the accompanying formal verification.

The fidelity-effect characterization is due to Rains [1]. Lami's quadratic converse already implies the half entropy-deficit upper bound [2]; we supply a direct finite-dimensional derivation and matching local constructions. Schur–Weyl universal compression and estimation are established methods [3, 4, 7]. Exact symmetric-state compression, including efficient qubit constructions, predates this work [14, 15]. We do not claim the symmetric-subspace dimension or compression mechanism as new. Classical source-coding reliability and second-order asymptotics are likewise prior results [6].

Recent universal distillation and coding results remove input-state knowledge under other free operations [8, 9, 10, 11]. State-agnostic work extraction studies a thermodynamic model [12, 13]. Their operations and resources differ from the global entangling preprocessing allowed here. Our claims concern the stated two-stage model: a common deterministic physical packing, matching completely PPT converses, finite-block control with visible dimension dependence, and a sharp target-uniform minimax law for an unknown rank-bounded support. The fact that a protocol is universal, by itself, is not presented as a new principle.

Purity-based extraction also includes the stochastic-unital analysis of Kondra et al. [25], which characterizes complete absolute separability, studies stability under adjoining maximally mixed local systems, and proves finite-copy activation. Their probabilistic convertibility problem permits postselection and compares states in the same output space. Our deterministic, variable-target fidelity optimization has a different resource specification. The qualitative possibility of activation is prior work.

2 Finite-dimensional spectral and subspace principles

Throughout this section the input dimensions are arbitrary integers $2 \le a \le b$, $N = ab$, and $K \ge 2$. The case $K = 1$ is the trivial exact product target.

Lemma 2.1 (Fidelity effects, following Rains)**.** *For a completely PPT channel let* $M = \Lambda^*(\Phi_K)$*. Then*

$$0 \le M \le I, \qquad -I/K \le M^\Gamma \le I/K. \tag{4}$$

Conversely every such M *is a fidelity effect of a completely PPT channel. Moreover,*

$$\operatorname{Tr} M^2 \le N/K^2, \qquad \|M\|_\infty \le \beta := \min(1, a/K). \tag{5}$$

Proof. The adjoint of a channel is positive and unital. Since $\Phi_K^\Gamma = F_K/K$ for the swap F_K, and $-I \le F_K \le I$, apply the adjoint of $\Gamma_{\text{out}}\Lambda\Gamma_{\text{in}}$ to these order bounds. This gives (4).

For the converse define

$$\Lambda_M(X) = \operatorname{Tr}(MX)\Phi_K + \operatorname{Tr}((I - M)X)\frac{I - \Phi_K}{K^2 - 1}.$$

This is a measure-and-prepare channel. Writing $Q_\pm = (I \pm F_K)/2$, its partial-transpose conjugate is

$$X \longmapsto \operatorname{Tr}\left[\frac{I + KM^\Gamma}{K(K+1)}X\right] Q_+ + \operatorname{Tr}\left[\frac{I - KM^\Gamma}{K(K-1)}X\right] Q_-.$$

Both coefficient effects are positive, and the map is trace preserving because $\operatorname{Tr} Q_\pm = K(K \pm 1)/2$. Its target overlap is $\operatorname{Tr}(MX)$.

Partial transpose preserves the Hilbert–Schmidt norm. The eigenvalues of M^Γ belong to $[-1/K, 1/K]$, proving the first bound in (5). For a unit vector v, the partial transpose of vv^* has trace norm $(\sum_j t_j)^2 \le a$, where t_j are the Schmidt coefficients. Hence

$$\langle v, Mv\rangle = \operatorname{Tr}\Big(M^\Gamma (vv^*)^\Gamma\Big) \le a/K.$$

Together with $M \le I$, this proves the norm bound. □

Lemma 2.2 (Coherent common-subspace packing)**.** *Let $S \subseteq \mathbb{C}^a \otimes \mathbb{C}^b$ be a known subspace of dimension r. If*

$$1 \le d \le \min(a, K), \qquad \lfloor a/d \rfloor \lfloor b/d \rfloor \ge r, \tag{6}$$

one unitary on the original input followed by deterministic product channels sends every density matrix supported on S to the same embedded Φ_d. Its target-K fidelity is d/K.

Proof. Put $u = \lfloor a/d \rfloor$ and $v = \lfloor b/d \rfloor$. In local coordinate blocks there are uv orthonormal vectors

$$|\psi_{ij}\rangle = d^{-1/2} \sum_{t=0}^{d-1} |id + t\rangle_A |jd + t\rangle_B, \quad 0 \le i < u, \quad 0 \le j < v.$$

Map a fixed orthonormal basis of S to r distinct such vectors and extend the resulting isometry to a unitary on the full N-dimensional input. Alice uses

$$A_i = \sum_{t<d} |t\rangle_K \langle id + t|,$$

and $|0\rangle\langle z|$ for each unused input coordinate z. Bob uses the corresponding maps. The Kraus adjoint products sum to I_a and I_b, including all unused coordinates.

For these product channels,

$$|\psi_{ij}\rangle\langle\psi_{i'j'}| \longmapsto \delta_{ii'}\delta_{jj'}\Phi_d.$$

Thus any logical matrix X on the selected code maps to $(\operatorname{Tr} X)\Phi_d$, including every off-diagonal coherence. A density matrix on S therefore has output Φ_d. Finally $\operatorname{Tr}(\Phi_K \Phi_d) = d/K$. No branch is postselected or renormalized. □

Proposition 2.3 (A subspace-versus-average comparison)**.** *Let a family $\mathscr{F}$ of input states be supported on one r-dimensional subspace, and let Ω be a probability average of states in that family. Define*

$$d_{a,b}(r) = \max\{d \in \{1, \dots, a\} : \lfloor a/d \rfloor \lfloor b/d \rfloor \ge r\}.$$

For the minimax fidelity over $\mathscr{F}$,

$$\min(1, d_{a,b}(r)/K) \le \sup_{U,\Lambda\in\mathrm{LO}} \inf_{\sigma\in\mathscr{F}} F(U,\Lambda;\sigma)$$
$$\le \sup_{U,\Lambda\in\mathrm{cPPT}} \inf_{\sigma\in\mathscr{F}} F(U,\Lambda;\sigma) \le \min\left(1, \frac{a}{K}, \frac{\sqrt{ab\,\mathrm{Tr}\,\Omega^2}}{K}\right). \tag{7}$$

The intermediate LOCC minimax value satisfies the same bounds.

Proof. Use Lemma 2.2 with $d = \min(K, d_{a,b}(r))$. For the upper bound, the infimum for a fixed protocol is at most its average,

$$\inf_{\sigma\in\mathscr{F}} F(U,\Lambda;\sigma) \le \mathrm{Tr}(MU\Omega U^*) \le \sqrt{\mathrm{Tr}\,M^2\,\mathrm{Tr}\,\Omega^2}.$$

Apply (5); the other two bounds follow from the trace-one condition and the norm bound. Averaging is used only to prove a converse, not as an operation in the protocol. □

We also need a spectral comparison for a single input. Let its eigenvalues be $\lambda_1 \ge \cdots \ge \lambda_N \ge 0$, and put

$$L_r = \sum_{i\le r} \lambda_i, \quad v_r = \min\left(1, \frac{a}{K}, \frac{\sqrt{N/r}}{K}\right), \quad T_K = \max_{1\le r\le N} v_r L_r. \tag{8}$$

Theorem 2.4 (One-shot spectral envelope)**.** *For every input spectrum and every $K \ge 2$,*

$$T_K/4 \le f^{\mathrm{LO}}(\lambda, K) \le f^{\mathrm{LOCC}}(\lambda, K) \le f^{\mathrm{cPPT}}(\lambda, K) \le H_N T_K, \qquad H_N = \sum_{i=1}^{N} \frac{1}{i}. \tag{9}$$

For $1 < \alpha \le 2$ one also has

$$f^{\mathrm{cPPT}}(\lambda, K) \le \beta^{(2-\alpha)/\alpha} (N/K^2)^{(\alpha-1)/\alpha} \left(\sum_i \lambda_i^\alpha\right)^{1/\alpha}. \tag{10}$$

At $\alpha = 1$ the bound is $f \le \beta$.

Proof. The decreasing eigenvalues μ_r of a fidelity effect satisfy $\mu_r \le v_r$, by (5). The spectral trace inequality and $\lambda_i \le L_i/i$ give

$$\mathrm{Tr}(MU\rho U^*) \le \sum_i \lambda_i \mu_i \le \sum_i \lambda_i v_i \le H_N T_K.$$

For a prefix rank r put $w = \min(a, K, \sqrt{N/r})$ and $d = \max(1, \lfloor w/2 \rfloor)$. Then $d \ge w/4$ and (6) holds. Indeed $d = 1$ is always feasible; otherwise $\lfloor a/d \rfloor \lfloor b/d \rfloor \ge N/(4d^2) \ge N/w^2 \ge r$. Apply Lemma 2.2 to the span of the top r eigenvectors. The remaining positive input contributes nonnegatively, giving fidelity at least $(d/K)L_r \ge v_r L_r/4$.

For (10) let $q = \alpha/(\alpha-1) \ge 2$. The effect eigenvalues satisfy $\mathrm{Tr}\,M^q \le \beta^{q-2}\,\mathrm{Tr}\,M^2$. Schatten Holder and (5) give the result. □

3 A common protocol and explicit finite-block bounds

3.1 Representation estimates

Use one fixed Schur–Weyl identification

$$(\mathbb{C}^D)^{\otimes k} = \bigoplus_{\lambda \vdash k,\, \ell(\lambda) \le D} Q_\lambda(D) \otimes P_\lambda, \qquad \rho^{\otimes k} = \bigoplus_\lambda Q_\lambda(\rho) \otimes I_{v_\lambda}. \tag{11}$$

Let $u_\lambda = \dim Q_\lambda(D)$, $v_\lambda = \dim P_\lambda$, $r_\lambda = u_\lambda v_\lambda$, and let J be the number of these diagrams. The identification and all bases are chosen independently of ρ. Schur–Weyl duality, irreducibility, the Weyl and hook dimension formulas, and the highest-weight character estimate are standard representation-theoretic inputs [4]. We use them in their usual finite-dimensional complex form, including singular positive inputs.

Pad λ by zeros to D coordinates. The dimension formulas are

$$u_\lambda = \prod_{i<j} \frac{\lambda_i - \lambda_j + j - i}{j - i}, \qquad v_\lambda = \frac{k! \prod_{i<j}(\lambda_i - \lambda_j + j - i)}{\prod_i (\lambda_i + D - i)!}. \tag{12}$$

Set $h_D = D(D-1)/2$, $Q = (k+D)^{h_D + D}$, and $q = \lambda/k$. Then

$$\begin{gathered} J \le (k+1)^D, \quad u_\lambda \le (k+D)^{h_D}, \\ Q^{-1} e^{kH(q)} \le v_\lambda \le e^{kH(q)}, \qquad Q^{-1} e^{kH(q)} \le r_\lambda \le Q e^{kH(q)}. \end{gathered} \tag{13}$$

For the bounds on v_λ, comparison of (12) with $k!/\prod_i \lambda_i!$ gives a ratio between $(k+D)^{-h_D}$ and one. For each i, compare the factors $\lambda_i - \lambda_j + j - i$ with $\lambda_i + j - i$, for $j > i$. The multinomial coefficient is between $(k+1)^{-D} e^{kH(q)}$ and $e^{kH(q)}$: the probability of its own type under q is a mode among at most $(k+1)^D$ types, and is at most one. This proves (13).

Let p be the decreasing eigenvalue law of ρ. The block probabilities satisfy

$$t_\lambda(p) = \operatorname{Tr}(\Pi_\lambda \rho^{\otimes k}) = v_\lambda s_\lambda(p), \qquad Q^{-1} e^{-kD(q\|p)} \le t_\lambda(p) \le Q e^{-kD(q\|p)}. \tag{14}$$

Indeed the highest-weight monomial p^λ occurs once, every other weight is dominated by λ, and the character has u_λ monomials counted with multiplicity. Thus $p^\lambda \le s_\lambda(p) \le u_\lambda p^\lambda$. If $\operatorname{rank} \rho = s < D$, diagrams with more than s rows have zero weight; all remaining bounds hold on the positive support of p.

3.2 A physical construction for every finite blocklength

Put $a_0 = \lfloor a/J \rfloor$, $b_0 = \lfloor b/J \rfloor$ and $N_0 = a_0 b_0$. If $a_0 = 0$, use a fixed product-output channel. Otherwise assign to each diagram a disjoint a_0-coordinate sector at Alice and a disjoint b_0-coordinate sector at Bob. In the Schur block select

$$S_\lambda = Q_\lambda(D) \otimes \operatorname{span}\{e_1, \dots, e_{v'_\lambda}\}, \quad v'_\lambda = \min(v_\lambda, \lfloor N_0/u_\lambda \rfloor), \quad r'_\lambda = u_\lambda v'_\lambda. \tag{15}$$

Empty selections are allowed. If $r'_\lambda > 0$, choose the largest $d_\lambda \le \min(K, a_0)$ with

$$\lfloor a_0/d_\lambda \rfloor \lfloor b_0/d_\lambda \rfloor \ge r'_\lambda.$$

The choice $d_\lambda = 1$ is feasible. Apply the coherent packing of Lemma 2.2 in each assigned sector. All selected images are orthogonal, and their total dimension is at most $JN_0 \le N$, so the selected isometry extends to one unitary on the original input. Use the sector-specific discard Kraus maps and add $|0\rangle\langle z|$ for every remaining local coordinate. This gives one pair of complete local channels.

Truncation in (15) affects only the permutation-multiplicity factor, which is the identity in (11). It retains exactly $\theta_\lambda = v'_\lambda/v_\lambda$ of the block mass for every unknown, potentially nondiagonal $Q_\lambda(\rho)$. There are no selected-unselected coherences in that factor. Hence

$$F_k^{\mathrm{u}}(\rho, K) \ge \sum_\lambda \theta_\lambda \frac{d_\lambda}{K} t_\lambda(p), \tag{16}$$

where empty selections contribute zero. In particular, if

$$K \le a_0, \qquad r_\lambda \le G := \lfloor a_0/K \rfloor \lfloor b_0/K \rfloor, \tag{17}$$

the full block is selected and $d_\lambda = K$. Therefore

$$F_k^{\mathrm{u}}(\rho, K) \ge \sum_{r_\lambda \le G} t_\lambda(p) \quad (K \le a_0). \tag{18}$$

The construction is deterministic, is defined for every finite k, and has quantifiers (2).

For fixed dimensions, eventually $a, b \ge 2J$ and $N_0 \ge 2 \max_\lambda u_\lambda$. Then

$$\theta_\lambda \ge \frac{1}{8J^2}, \qquad \frac{d_\lambda}{K} \ge \frac{1}{8J} \min\left(1, \frac{a}{K}, \frac{\sqrt{N/r_\lambda}}{K}\right).$$

For the first inequality, either no truncation occurs or $\lfloor N_0/u_\lambda \rfloor \ge N_0/(2u_\lambda)$, while $N_0 \ge N/(4J^2)$. For the second, use the feasible choice $\max(1, \lfloor w/2 \rfloor)$ with $w = \min(K, a_0, \sqrt{N_0/r'_\lambda})$, which is at least $w/4$. Thus the same protocol obeys

$$F_k^{\mathrm{u}}(\rho, K) \ge \frac{1}{64J^3} \max_\lambda \min\left(1, \frac{a}{K}, \frac{\sqrt{N/r_\lambda}}{K}\right) t_\lambda(p). \tag{19}$$

This is a bound for one common protocol, not a different unitary for each maximizing diagram.

3.3 Information-spectrum sandwich

Sample X with law p on its positive support, and set $Z = -\log p_X$ and $S_k = Z_1 + \cdots + Z_k$. These variables describe performance; they are not supplied to the protocol. Let $u_* = \max_\lambda u_\lambda$.

Theorem 3.1 (Dimension-explicit finite-block sandwich)**.** *For the common protocol above, whenever $K \le a_0$,*

$$\mathbb{P}\{S_k \le \log(G/u_*)\} \le F_k^{\mathrm{u}}(\rho, K). \tag{20}$$

Every state-aware completely PPT protocol, for every $t > 0$, satisfies

$$F_k^{\star}(\rho, K) \le \mathbb{P}\{S_k \le \log(N/K^2) + t\} + e^{-t/2}. \tag{21}$$

If also $a \geq 2J$, put

$$x = \log(N/K^2), \qquad h = \log(16J^2 u_*).$$

Then

$$\mathbb{P}\{S_k \leq x - h\} \leq F_k^{\mathrm{u}}(\rho, K) \leq F_k^{\star}(\rho, K) \leq \mathbb{P}\{S_k \leq x + t\} + e^{-t/2}, \tag{22}$$

simultaneously for all finite dimensions and all input states, including zeros in their spectra.

Proof. On each Schur block, $v_\lambda \operatorname{Tr} Q_\lambda(\rho) = t_\lambda \leq 1$. Therefore $Q_\lambda(\rho) \leq I/v_\lambda$. Every positive input eigenvalue z in a bad block $r_\lambda > G$ satisfies

$$z \leq \frac{1}{v_\lambda} = \frac{u_\lambda}{r_\lambda} < \frac{u_*}{G}.$$

The event in (20) can thus occur only in good blocks; apply (18). This uses the input self-information, rather than a presumed central limit theorem for diagrams. The same block norm estimate underlies standard universal-state domination bounds [7].

For (21), absorb the fixed input unitary into the effect and take its diagonal entries z_i in an eigenbasis of $\rho^{\otimes k}$. They satisfy $0 \leq z_i \leq 1$ and $\sum_i z_i^2 \leq \operatorname{Tr} M^2 \leq r = N/K^2$. Eigenvalues $p_i^{(k)} < e^{-(\log r + t)}$ contribute at most

$$\sqrt{\Big(\sum_{\text{small}} (p_i^{(k)})^2\Big)\Big(\sum_{\text{small}} z_i^2\Big)} \leq \sqrt{e^{-(\log r + t)} r} = e^{-t/2}.$$

The other contribution is at most the probability in (21). Finally $a_0 \geq a/(2J)$, $b_0 \geq b/(2J)$, and $a_0, b_0 \geq K$ imply $G \geq N/(16J^2K^2)$, giving (22). □

Corollary 3.2 (A quantitative Gaussian bound)**.** *Let $V = \operatorname{Var} Z > 0$, $T = \mathbb{E}|Z - H(p)|^3$, $\sigma = \sqrt{kV}$, and*

$$\beta = \frac{C_{\mathrm{BE}} T}{V^{3/2}\sqrt{k}}, \qquad y = \frac{\log(N/K^2) - kH(p)}{\sigma},$$

where C_{BE} is any valid absolute iid Berry–Esseen constant. Under the hypotheses of (22)*,*

$$\Phi(y - h/\sigma) - \beta \leq F_k^{\mathrm{u}}(\rho, K), \qquad F_k^{\star}(\rho, K) \leq \Phi(y + t/\sigma) + \beta + e^{-t/2}.$$

In particular, for $\sigma > 1$, each fidelity differs from $\Phi(y)$ by at most

$$\beta + \frac{\max(h, 2\log\sigma)}{\sqrt{2\pi}\,\sigma} + \frac{1}{\sigma}. \tag{23}$$

Proof. Apply the ordinary iid Berry–Esseen inequality [18] to S_k, use the Lipschitz constant $1/\sqrt{2\pi}$ of Φ, and set $t = 2\log\sigma$. Both fidelity bounds hold because $F_k^{\mathrm{u}} \leq F_k^{\star}$. □

This estimate applies row by row to growing dimensions and changing states. Sufficient conditions for a common Gaussian approximation are

$$kV \to \infty, \qquad \frac{T}{V^{3/2}\sqrt{k}} \to 0, \qquad \frac{h}{\sqrt{kV}} \to 0, \tag{24}$$

in addition to $a \geq 2J$ and $K \leq a_0$. For instance

$$h \leq \log 16 + 2D \log(k+1) + \frac{D(D-1)}{2} \log(k+D).$$

If V is bounded above and below by positive constants times $1/D$ and $T = O(1/D)$, then $D^5 \log^2(k+D) = o(k)$ is sufficient for (24). This is a conservative sufficient condition for this construction; no necessity or automatic APPT moment hypothesis is asserted.

4 Exact asymptotic extraction laws

The limits in this section fix m, n and an input state ρ. Let p be its eigenvalue law, let $s = \operatorname{rank} \rho$, and put

$$A = \log m, \quad L = \log D, \quad C(p) = \min\left(A, \frac{L - H(p)}{2}\right), \quad C_0(p) = \min\left(A, \frac{L - \log s}{2}\right).$$

Write $S_\alpha(p) = (1-\alpha)^{-1} \log \sum_i p_i^\alpha$, with $S_1 = H$, and take $K_k = \lceil e^{kR} \rceil$ for a fixed $R \geq 0$. At $R = 0$ the target is one dimensional and fidelity is exactly one.

Theorem 4.1 (One all-state family with the optimal laws)**.** *The single family constructed in Section 3 has the following properties for every fixed input state, including rank-deficient states.*

1. *The faithful extraction capacities for* LO, LOCC, *and* cPPT *all equal* $C(p)$*. The common family achieves fidelity tending to one for each* $R < C(p)$*.*

2. *At every rate, its fidelity exponent equals that of the state-aware optimum in each of the three operation classes:*

$$\lim_{k\to\infty} -\frac{1}{k} \log F_k^{\mathrm{u}}(\rho, K_k) = \lim_{k\to\infty} -\frac{1}{k} \log f_k^{\mathcal{C}}(\rho, K_k) = E_p(R), \tag{25}$$

   where

$$E_p(R) = (R - A)_+ + \max_{1 \leq \alpha \leq 2} \frac{\alpha - 1}{\alpha} \big(S_\alpha(p) - c_R\big), \qquad c_R = L - 2\min(R, A). \tag{26}$$

   The summand at $\alpha = 1$ *is zero. In particular* $R > C(p)$ *gives a strictly positive strong-converse exponent.*

3. *In the strict direct regime* $C_0(p) < R < C(p)$*, put* $c = L - 2R$*. Then*

$$\begin{aligned} \lim_{k\to\infty} -\frac{1}{k} \log(1 - F_k^{\mathrm{u}}) = \lim_{k\to\infty} -\frac{1}{k} \log(1 - F_k^{\star}) = I_p(c), \\ I_p(c) := \min_{H(q) \geq c} D(q\|p) = \sup_{0<\alpha<1} \frac{1-\alpha}{\alpha}(c - S_\alpha(p)). \end{aligned} \tag{27}$$

   Relative entropy is evaluated on the positive support of p*.*

4. *For $R < C_0(p)$ the same universal protocol has fidelity exactly one for all sufficiently large k. The eventual-exact capacity is $C_0(p)$, even for state-aware completely PPT postprocessing.*

Proof of the fidelity exponent. The converse follows from (10) on tensor powers and $k^{-1}\log K_k \to R$. For achievability, approximate a decreasing distribution q on the positive support of p by diagrams $\lambda^{(k)}/k$. Equations (13), (14), and (19) imply

$$\limsup_k -\frac{1}{k}\log F_k^{\mathrm{u}} \le \min_q \left\{D(q\|p) + \max\left(0, R-A, R-L/2+H(q)/2\right)\right\}. \tag{28}$$

Every omitted factor has logarithm $O_{m,n}(\log k)$. Sorting q cannot increase $D(q\|p)$, by rearrangement with decreasing p, and leaves its entropy unchanged; hence decreasing diagrams lose no minimizer. The maximum in (28) is $(R-A)_+ + (H(q)-c_R)_+/2$.

For completeness the matching scalar identity, with no exchange of minimax operations, is

$$\min_q\{D(q\|p) + (H(q)-c)_+/2\} = \max_{1\le\alpha\le 2}\frac{\alpha-1}{\alpha}(S_\alpha(p)-c), \qquad c \ge 0. \tag{29}$$

Let $q_{\alpha,i} = p_i^\alpha/Z_\alpha$, $Z_\alpha = \sum_i p_i^\alpha$, and call the maximand $J_c(\alpha)$. Direct differentiation gives

$$J_c'(\alpha) = \frac{H(q_\alpha)-c}{\alpha^2}, \qquad \frac{d}{d\alpha}H(q_\alpha) = -\alpha\,\mathrm{Var}_{q_\alpha}(\log p_i) \le 0.$$

If $H(p) \le c$, use $q = p$, and both sides vanish. If $H(q_2) \ge c$, use $q = q_2$, giving $(S_2(p)-c)/2$. Otherwise an escort with $1 < \alpha < 2$ has $H(q_\alpha) = c$, and its relative entropy is $J_c(\alpha)$. Boundary cases agree. Conversely, for every q and $1 \le \alpha \le 2$,

$$D(q\|p) = \alpha^{-1}D(q\|q_\alpha) + \frac{\alpha-1}{\alpha}(S_\alpha(p) - H(q)),$$

while $(H-c)_+/2 \ge (\alpha-1)(H-c)/\alpha$. This proves (29). Since $c_R \ge \log(n/m) \ge 0$, the identity applies and proves (25) by squeezing all three operation classes between the common LO protocol and the cPPT converse. □

Proof of faithful and eventual-exact capacities. For $0 < R < C(p)$ one has $R < A$, so eventually $K_k \le a_0$. In (22), $x - h = k(L-2R) + O(\log k) > k(H(p)+\delta)$ for a fixed $\delta > 0$. The ordinary law of large numbers implies fidelity tending to one. Rate zero is trivial. Above $C(p)$ the exponent in (26) is positive: either $R > A$, or $J'_{L-2R}(1) > 0$.

If $R < C_0(p)$, every block of nonzero probability has at most s rows and $r_\lambda \le Qs^k$. On the other hand,

$$G \ge \frac{N}{16J^2K_k^2} \ge \frac{e^{k(L-2R)}}{64J^2}$$

eventually. Since $L - 2R > \log s$, all nonzero blocks are good and (18) equals one. Conversely, exact fidelity for a rank-s^k state forces its rotated support into the eigenvalue-one space of M. Thus $s^k \le \mathrm{Tr}\,M^2 \le N/K^2$ and $K \le a$. These bounds give the claimed eventual-exact capacity. □

Proof of strict-direct reliability. The hypotheses imply $H(p) < c < \log s$. If a block fails (17), the bounds above give

$$H(\lambda/k) > c - \frac{\log(64J^2Q)}{k}.$$

Using (14) and at most polynomially many diagrams yields the lower bound $I_p(c)$ on the error exponent of the common protocol.

To prove optimality, fix $0 < t < 1$. At most $N/(K^2t^2)$ eigenvalues of M exceed t. If L_j is the mass of the largest j input eigenvalues, capped at one beyond their support, the spectral trace inequality gives

$$1 - F_k^\star \geq (1-t)\left(1 - L_{\lfloor N/(K^2t^2)\rfloor}(p^{\otimes k})\right). \tag{30}$$

For integers M_k with $k^{-1}\log M_k \to c \in (H(p), \log s)$, the optimal-prefix tail has exponent $I_p(c)$. Here is the needed type argument. Include all types with entropy at most $c - \delta$; their total cardinality is at most $(k+1)^s e^{k(c-\delta)} \leq M_k$ eventually. The excluded mass is at most $(k+1)^s e^{-kI_p(c-\delta)}$. Conversely, a type approaching any q with $H(q) > c$ has exponentially more words than M_k, and at least fraction $1 - M_k/r_q$ of its equiprobable words must be excluded. Its mass is at least $(k+1)^{-s}e^{-kD(q\|p)}$. Approximate an entropy-c minimizer and let $\delta \downarrow 0$. With $t = 1/2$, (30) gives the matching upper bound on every error exponent.

The requisite continuity and the last expression in (27) follow from escorts with $0 < \alpha < 1$. Since $H(p) < c < \log s$, the positive eigenvalues are nonuniform and a unique escort has entropy c. The identity

$$D(q\|p) = \alpha^{-1}D(q\|q_\alpha) + \frac{1-\alpha}{\alpha}(H(q) - S_\alpha(p))$$

proves the variational formula, equality at that escort, and continuity in the strict interval. □

Remark 4.2 (The exact zero-error boundary). Formula (27) excludes $R = C_0(p)$. For a rank-four state on 4×4, $K_k = 2^k$ permits exact packing of its 4^k-dimensional support into Φ_{2^k} and garbage. Thus the state-aware error can be exactly zero at $R = \log 2 = C_0(p)$ even for nonuniform positive eigenvalues. A finite reliability expression at $c = \log 4$ would miss this integer packing effect.

Theorem 4.3 (Entropy-limited Gaussian threshold)**.** *Suppose* $0 < C(p) < A$ *and*

$$V(p) = \sum_{p_i>0} p_i(-\log p_i - H(p))^2 > 0.$$

For any target sequence with $\log K_k = kC(p) + z\sqrt{k} + o(\sqrt{k})$,

$$F_k^{\mathrm{u}}(\rho, K_k) \longrightarrow \Phi(-2z/\sqrt{V(p)}), \qquad F_k^\star(\rho, K_k) \longrightarrow \Phi(-2z/\sqrt{V(p)}). \tag{31}$$

For fixed $0 < \varepsilon < 1$*, the state-aware optimal fixed-error target obeys*

$$\log K_k^{\mathrm{opt}}(\varepsilon) = kC(p) + \tfrac{1}{2}\sqrt{kV(p)}\,\Phi^{-1}(\varepsilon) + o(\sqrt{k}),$$

and the common family attains the same expansion.

Proof. The strict local-dimension gap ensures $K_k \le a_0$ eventually. Here $h = O(\log k)$ and $\log(N/K_k^2) = kH(p) - 2z\sqrt{k} + o(\sqrt{k})$. Apply the ordinary central limit theorem to the two sides of (22), for example with $t = k^{1/4}$; alternatively use Corollary 3.2. To obtain the fixed-error expansion, bracket each requested integer target above and below a fixed Gaussian quantile. The converse applies to every larger target, and the construction applies to the chosen smaller target, so no monotonicity of the particular universal family is needed. □

Theorem 4.3 retains both strict hypotheses. The local-dimension cap and zero varentropy are not included by continuity. The next section treats a cap-boundary phenomenon of a different, minimax kind.

5 The sharp cost of an unknown support

5.1 A common support and an adversarial average

Fix $1 \le s \le D$. Define the known subspace

$$S_{k,s} = \bigoplus_{\ell(\lambda)\le s} Q_\lambda(D) \otimes P_\lambda, \qquad r_{k,s} = \sum_{\ell(\lambda)\le s} u_\lambda(D) v_\lambda. \tag{32}$$

Every $\rho^{\otimes k}$ with $\operatorname{rank}\rho \le s$ is supported on $S_{k,s}$. Let P_s be one fixed rank-s projection on $\mathbb{C}^D$, and form the mathematical average

$$\Omega_{k,s} = \int_{W\in U(D)} (W P_s W^*/s)^{\otimes k}\, dW. \tag{33}$$

The normalized Haar measure is used only to construct an adversarial input ensemble; it is never implemented as preprocessing.

Schur's lemma and the character identity give, on a block with at most s rows, the eigenvalue

$$\frac{u_\lambda(s)}{s^k u_\lambda(D)} \quad \text{with multiplicity } u_\lambda(D)v_\lambda. \tag{34}$$

Indeed the trace of $Q_\lambda(P_s)$ is $s_\lambda(1,\dots,1,0,\dots,0) = u_\lambda(s)$; averaging the irreducible representation divides this trace by $u_\lambda(D)$. All these eigenvalues are positive. Thus $S_{k,s}$ is exactly the smallest common support subspace of the rank-budget class.

Theorem 5.1 (Support dimension and reciprocal average purity)**.** *For fixed D and $1 \le s < D$, put*

$$g = s(D-s), \qquad c_{D,s} = \frac{s^{-g}}{\displaystyle\prod_{1\le i\le s<j\le D} (j-i)}. \tag{35}$$

Then

$$r_{k,s} \sim c_{D,s} s^k k^g, \qquad \operatorname{Tr}\Omega_{k,s}^2 \sim (c_{D,s} s^k k^g)^{-1}, \qquad r_{k,s}\operatorname{Tr}\Omega_{k,s}^2 \longrightarrow 1. \tag{36}$$

Proof. Cancel the factors in the Weyl products to obtain, for $\ell(\lambda) \le s$,

$$R_\lambda := \frac{u_\lambda(D)}{u_\lambda(s)} = \prod_{1\le i\le s<j\le D} \frac{\lambda_i + j - i}{j-i} \ge 1. \tag{37}$$

The Schur–Weyl probability law for I_s/s is

$$t_\lambda = \frac{u_\lambda(s) v_\lambda}{s^k}, \qquad \sum_\lambda t_\lambda = 1.$$

Equations (32) and (34) now give the exact two-moment identities

$$r_{k,s} = s^k \mathbb{E}_t R_\lambda, \qquad \operatorname{Tr} \Omega_{k,s}^2 = s^{-k} \mathbb{E}_t (R_\lambda^{-1}). \tag{38}$$

Under t, λ/k concentrates exponentially near $(1/s, \dots, 1/s)$. To see this directly, the character and dimension bounds yield

$$t_\lambda \le (k+s)^{s(s-1)/2} \exp\{-kD(\lambda/k \| I_s/s)\}.$$

There are at most $(k+1)^s$ diagrams. Off any fixed neighborhood of the uniform vector, compactness gives a strictly positive lower bound on the relative entropy. The probability there is consequently bounded by a polynomial times e^{-ck}.

The ratio R_λ/k^g is uniformly bounded for $k \ge 1$. Near the uniform vector, its product formula tends to $c_{D,s}$. Hence $k^{-g}\mathbb{E}R_\lambda \to c_{D,s}$. For the inverse moment fix a sufficiently small neighborhood in which $\lambda_i \ge k/(2s)$ for every $i \le s$. On this neighborhood, k^g/R_λ is uniformly bounded and tends to $1/c_{D,s}$ as the neighborhood contracts. On its complement, $R_\lambda \ge 1$ bounds k^g/R_λ by k^g; its expectation there tends to zero by the exponential tail. This supplies the required uniform integrability, and proves $k^g \mathbb{E} R_\lambda^{-1} \to 1/c_{D,s}$. Substitute into (38). □

At $s = 1$, these identities reduce to

$$S_{k,1} = \operatorname{Sym}^k(\mathbb{C}^D), \qquad r_{k,1} = \binom{k+D-1}{D-1}, \qquad \Omega_{k,1} = \Pi_{\mathrm{sym}}/r_{k,1}.$$

At $s = D$ one instead has exactly $r_{k,D} = N$ and $\Omega_{k,D} = I/N$.

5.2 A target-uniform minimax law

Define the finite, explicitly computable quantities

$$d_k = d_{a,b}(r_{k,s}), \qquad w_k = \min\left(a, \sqrt{N/r_{k,s}}\right), \qquad \eta_k = \sqrt{r_{k,s} \operatorname{Tr} \Omega_{k,s}^2}. \tag{39}$$

Proposition 2.3 gives for every finite k and every integer $K \ge 1$

$$\begin{aligned} &\min(1, d_k/K) \le \mathcal{G}_{\mathrm{LO}}(k, K; s) \le \mathcal{G}_{\mathrm{LOCC}}(k, K; s) \le \mathcal{G}_{\mathrm{cPPT}}(k, K; s), \\ &\mathcal{G}_{\mathrm{cPPT}}(k, K; s) \le \min\left(1, \frac{a}{K}, \frac{\sqrt{N \operatorname{Tr} \Omega_{k,s}^2}}{K}\right). \end{aligned} \tag{40}$$

The lower construction knows s, but no property of the particular input within the rank-budget class.

Theorem 5.2 (Entire sharp minimax profile)**.** *For every fixed $2 \le m \le n$ and $1 \le s < D$, and every $\mathcal{C} \in \{\mathrm{LO}, \mathrm{LOCC}, \mathrm{cPPT}\}$,*

$$\sup_{K \in \mathbb{N},\ K \ge 1} \left| \frac{\mathcal{G}_{\mathcal{C}}(k, K; s)}{\min(1, w_k/K)} - 1 \right| \longrightarrow 0. \tag{41}$$

An explicit finite-k bound for the supremum is

$$\max\{1 - d_k/w_k, \eta_k - 1\}. \tag{42}$$

Thus the relative equivalence includes every target integer, even when fidelity is exponentially small.

Proof. It remains to prove $d_k/w_k \to 1$, since $\eta_k \to 1$ by Theorem 5.1. If $n > ms$, then $r_{k,s}/(b/a) \to 0$ and eventually $d = a$ is feasible. Consequently $d_k = w_k = a$ eventually.

If $n \le ms$, Theorem 5.1 gives eventually

$$w_k = \sqrt{N/r_{k,s}}, \qquad w_k/a \to 0, \quad w_k/b \to 0, \quad w_k \to \infty.$$

The polynomial factor is essential when $n = ms$. Put $q_k = w_k/a + w_k/b$, $\epsilon_k = \sqrt{q_k}$, and $d = \lfloor w_k/(1+\epsilon_k) \rfloor$. For large k this is a positive integer with $d/w_k \to 1$. Dividing a lower bound for the product of floors by $r = N/w_k^2$ gives

$$\frac{\lfloor a/d \rfloor \lfloor b/d \rfloor}{r} \ge (w_k/d)^2 - (w_k/d) q_k + 1/r.$$

For $0 < q_k < 1$ the function $z^2 - z q_k$ is increasing on $z \ge 1$, and $w_k/d \ge 1 + \sqrt{q_k}$. The last display is therefore at least $1 + 2\sqrt{q_k} - q_k^{3/2} > 1$. Thus d is feasible. Also every feasible integer is at most w_k, so $d_k/w_k \to 1$.

Finally (40) bounds the ratio in (41) between d_k/w_k and η_k, independently of K. This proves both assertions. □

For $s = D$ the exact minimax value is $1/K$ at every blocklength: the maximally mixed state gives this upper bound and a product output attains it.

Corollary 5.3 (Exact and fixed-error target sizes)**.** *Let $K^0_{\mathcal{C}}(k; s)$ be the largest target dimension admitting exact fidelity for every state in the rank-budget class. Then*

$$d_k \le K^0_{\mathrm{LO}} \le K^0_{\mathrm{LOCC}} \le K^0_{\mathrm{cPPT}} \le \lfloor w_k \rfloor. \tag{43}$$

If $n \le ms$ and $s < D$,

$$\log K^0_{\mathcal{C}}(k; s) = \frac{k}{2} \log(D/s) - \frac{s(D-s)}{2} \log k - \frac{1}{2} \log c_{D,s} + o(1). \tag{44}$$

If $n > ms$, instead $K^0_{\mathcal{C}}(k; s) = m^k$ eventually. For fixed $0 < \varepsilon < 1$, the largest target admitting worst-class fidelity at least $1 - \varepsilon$ satisfies

$$\log K^{\varepsilon}_{\mathcal{C}}(k; s) = \log w_k - \log(1 - \varepsilon) + o(1). \tag{45}$$

Proof. Exact performance gives $\mathrm{Tr}[\Omega_{k,s}(I - U^*MU)] = 0$. Positivity and strict positivity of $\Omega_{k,s}$ on $S_{k,s}$ force U^*MU to be the identity on that subspace. Hence $r_{k,s} \le \mathrm{Tr}\, M^2 \le N/K^2$ and $K \le a$. This proves the upper bound in (43); packing proves its lower bound. Use $d_k/w_k \to 1$ and (36) for (44). For fixed error, the actual lower construction works at $K = \lfloor d_k/(1-\varepsilon) \rfloor$, while (40) bounds every feasible target by $\eta_k w_k/(1-\varepsilon)$. Since $w_k \to \infty$ for $s < D$, integer rounding contributes $o(1)$ to the logarithm. □

The boundary $n = ms$ has first-order capacity $\log m$, but still incurs the logarithmic correction in (44). The coefficient $s(D-s)/2$ is one quarter of the real dimension of the Grassmannian of s-dimensional supports. This interpretation follows from the two-moment proof; parameter counting is not used as a substitute for it.

Corollary 5.4 (A known spectrum with unknown orientation)**.** *Fix any positive rank-s spectrum. Requiring one exact protocol for its entire global-unitary orbit gives the same bounds* (43) *and hence the same exact-target asymptotic. If the known spectrum is flat on rank s, the full target-uniform minimax profile* (41) *also holds for that orbit alone.*

Proof. For any positive rank-s spectrum the orbit average has a strictly positive coefficient on every Schur block with at most s rows, and zero elsewhere. Its support is exactly $S_{k,s}$. The exact converse and common-subspace packing are unchanged. For the flat spectrum its average is precisely (33), so the entire previous minimax argument applies. No fixed-error profile for an arbitrary nonuniform spectrum is inferred from this support statement. □

5.3 Balanced unknown pure inputs and an exact circuit

When $m = n$, $D = m^2$, and $s = 1$, put $r_k = \binom{k+D-1}{D-1}$. Theorem 5.2 gives, uniformly in every target integer,

$$\mathcal{G}_{\mathcal{C}}(k, K; 1) \sim \min\left(1, \frac{m^k}{K\sqrt{r_k}}\right).$$

In particular

$$\mathcal{G}_{\mathcal{C}}(k, m^k; 1) \sim r_k^{-1/2} \sim \sqrt{(D-1)!}\, k^{-(D-1)/2}. \tag{46}$$

A known pure state admits fidelity one at this same target. The common-protocol loss is therefore sharp and invisible to the first-order capacity.

The symmetric compression mechanism is established prior work [14, 15, 16]; coherent streaming concentration also predates this work [17]. The following specialization records its exact resource conversion in our model.

Proposition 5.5 (Optimal integer Bell pairs with no supplied memory)**.** *For $m = n = 2$ and every $k \ge 1$, one circuit on the original $2k$ qubits, followed by local discarding, extracts exactly*

$$h_k = \left\lfloor k - \frac{1}{2}\log_2 \binom{k+3}{3} \right\rfloor$$

Bell pairs from every unknown pure iid input. This integer is optimal even for a common global unitary followed by completely PPT postprocessing. The circuit can be synthesized with $O(k^7 \log^2(k+3))$ CNOT and arbitrary single-qubit gates, with no qubit supplied, reset,

measured, or discarded before the global circuit. A fixed efficiently computable inverse-closed universal one-qubit gate set with CNOT gives operator-norm approximation, up to global phase, within δ using an additional $O(\log^c(G/\delta))$ factor for a Solovay–Kitaev exponent c, where $G = O(k^7 \log^2(k+3))$. The output fidelity is at least $1 - 2\delta$.

Proof. Let $r_j = \binom{j+3}{3}$ and $\ell_j = \lceil \log_4 r_j \rceil$. Fix a computable enumeration enc_j of the four-component occupations of j. Sequentially compress the first j four-level systems into ℓ_j of those same systems; all other processed systems become zero on the entire symmetric subspace. This is possible because $r_j/r_{j-1} = (j+3)/j \le 4$, so $\ell_j \le \ell_{j-1} + 1 \le j$.

At step j, the only active systems are the ℓ_{j-1} memory systems and the next original input. In their computational basis the symmetric branching vectors are

$$v_\nu = \sum_{i:\nu_i>0} \sqrt{\nu_i/j}\,|\,\mathrm{enc}_{j-1}(\nu - e_i)\rangle|i\rangle.$$

Their computational supports are disjoint, and each has at most four entries. At most three real Givens rotations per occupation send it to one selected basis state. Complete these states to the desired $\mathrm{enc}_j(\nu)$ labels by a fixed permutation of the full active basis. If the active register has $q_j = 2(\ell_{j-1}+1)$ qubits and dimension $L_j = 2^{q_j}$, at most $4L_j$ two-level operations suffice, and $L_j \le 16 r_{j-1} = O(j^3)$. All rotations and the permutation are computable from polynomially many occupation labels, not from the input state.

For clarity an elementary ancilla-free synthesis bound for a two-level operation on q qubits is $O(q^2 2^q)$. A Gray path reduces it to pattern-controlled one-qubit rotations and phases. A phase on one selected string is $\exp(i\theta P_x)$ with

$$P_x = 2^{-q} \prod_l (I + (-1)^{x_l} Z_l).$$

Expand into at most 2^q commuting Pauli strings; implement each nonempty string by a parity circuit, a one-qubit rotation, and the inverse parity circuit, using an existing participating wire. Controlled Y rotations have the same expansion after a local change of basis. A Gray path uses at most $2q$ selective transpositions. Thus a conservative total bound is

$$1000 \sum_{j=2}^{k} q_j^2 L_j^2 + 100k^2 = O(k^7 \log^2(k+3)).$$

This agrees with the general synthesis context [19, 20], while making the absence of a supplied workspace explicit.

After compression at $j = k$, at most $\lceil \log_2 r_k \rceil \le 2(k - h_k)$ memory qubits can be nonzero. Redistribute them among the two garbage registers of $k - h_k$ qubits each. All remaining original qubits are already zero. Hadamards and cross-party CNOTs on the remaining h_k pairs create the desired Bell states; local discarding removes the garbage. The zero wires were generated from the promised input by the unitary compression, not supplied externally.

For necessity, exact fidelity for every pure input makes the effect the identity on the whole symmetric subspace. Lemma 2.1 then gives $r_k \le 4^k/4^h$, proving the exact integer optimum.

The elementary angles are square roots and inverse trigonometric functions of rational occupation ratios. They can be computed to accuracy $O(\delta/G)$ in time polynomial in k

and $\log(1/\delta)$. The Solovay–Kitaev theorem [21] compiles the rotations into the stated finite library. A telescoping unitary error bound, after choosing the irrelevant overall phase, gives trace-norm output error at most 2δ. Trace-norm contraction under local channels proves the fidelity claim. □

This proposition implements the pure-promise task. It does not provide an efficient implementation of the complete all-state mixed-input family of Section 3.

5.4 Compatibility with an all-state family

The sharp balanced pure profile can also be attained by a family retaining every pointwise mixed-state law of Section 4. To see this, put $q = (k+1)J$ and give each nonsymmetric diagram sectors of sizes $\lfloor a/q \rfloor$ and $\lfloor b/q \rfloor$. Give the symmetric block all remaining coordinates at each party. Apply the same multiplicity-only selection and maximal feasible Bell packing in each sector.

The nonsymmetric blocks still retain at least an inverse-polynomial fraction of their mass, with exponent loss at most $\log(64q^3) = O_D(\log k)$. Blocks below the corresponding good-rank threshold are still extracted exactly; the information-spectrum overhead is at most $\log(16q^2u_*) = O_D(\log k)$. The capacity, exact exponent, strict reliability, eventual-exact rate, and entropy-limited Gaussian theorem are consequently unchanged.

For a pure input only the symmetric block occurs. In balanced dimensions its allocated local dimension a_{sym} satisfies $a_{\mathrm{sym}}/a \to 1$. Packing this whole block gives fidelity at least

$$\min\left(1, \frac{\lfloor a_{\mathrm{sym}}/\lceil\sqrt{r_k}\rceil\rfloor}{K}\right),$$

which matches (46) and its entire target-uniform profile. Thus the pure-state sharp law need not require a purity flag. The general rank-s sharp minimax theorem, however, retains its stated rank-budget promise; this modification does not establish simultaneous sharp minimax constants for every $s > 1$.

6 Applications, verification, and open questions

6.1 APPT entropy as an extraction budget

A state on $\mathbb{C}^m \otimes \mathbb{C}^n$ is absolutely PPT if every global unitary conjugate has positive partial transpose. The capacity formula connects its spectral entropy to a collective extraction resource, even though no single-copy global rotation can produce an NPT state. Qualitative activation from purity is known [25]; the following identity records the quantitative link needed here.

Let

$$E_1(m,n) = \log D - \min_{\rho \in \mathrm{APPT}(m,n)} S(\rho).$$

The APPT set is compact: it is the intersection, inside the compact set of density matrices, of closed partial-transpose positivity conditions. Von Neumann entropy is continuous, so the

minimum is attained. Theorem 4.1 immediately gives

$$\max_{\rho\in\mathrm{APPT}(m,n)} C(p_\rho) = \min\left(\log m, \frac{E_1(m,n)}{2}\right). \tag{47}$$

Thus a minimum-entropy theorem for APPT spectra calibrates the maximal capacity in the present model. A companion all-dimension entropy profile can be substituted in this identity [28]; its proof and its verification status are separate from the present paper.

One can obtain a self-contained operational comparison from the already verified purity asymptotic [29]. For $P_{\max}(m,n) = \max_{\mathrm{APPT}} \operatorname{Tr}\rho^2$, Jensen's inequality gives $S(\rho) \geq -\log \operatorname{Tr}\rho^2$, and hence

$$\max_{\mathrm{APPT}} C \leq \frac{1}{2}\log(DP_{\max}). \tag{48}$$

For balanced systems $D = m^2 \to \infty$, the purity law $P_{\max} = D^{-1} + 8D^{-2} + o(D^{-2})$ yields $\max_{\mathrm{APPT}} C \leq 4/D + o(D^{-1})$.

For a lower example take $\rho = (I + 2vv^*)/(D+2)$. It is APPT for every unit vector v: after every global rotation, the smallest eigenvalue of $(vv^*)^\Gamma$ is at least $-1/2$, by the Schmidt coefficient inequality $t_i t_j \leq 1/2$. Its spectrum is $(3, 1, \ldots, 1)/(D+2)$, so for balanced sufficiently large dimensions

$$C(p_\rho) = \frac{1}{2}\left[\log\frac{D}{D+2} + \frac{3\log 3}{D+2}\right] = \frac{3\log 3 - 2}{2D} + O(D^{-2}).$$

These comparisons are applications of spectral and capacity results, not new claims about ordinary fixed-state LOCC distillation.

Equation (47) fixes the dimensions before taking the many-copy capacity limit. A joint growing-dimension conclusion requires uniform finite-block control. Theorem 3.1 supplies such control without spectral assumptions; a Gaussian conclusion additionally requires the moment and output-size conditions in (24). No assertion that every APPT near optimizer satisfies those conditions is made.

6.2 Scope of machine verification

The accompanying Lean development checks the actual finite-dimensional matrix and channel core of the method. Its inputs are complex matrices, genuine Hermitian projections, finite Kraus families with completeness, and positivity of the partial-transpose-conjugated map. The bounds are derived from those objects; the quadratic or dimension budgets are not supplied as extra assumptions.

The core package proves the following statement: if a projection P has $\operatorname{rank} P \leq \lfloor a/d\rfloor\lfloor b/d\rfloor$ and $0 < d \leq K$, there exist a fixed unitary on the original input and two complete local Kraus channels such that every positive trace-one matrix supported on P has target overlap exactly d/K. It includes projection factorization, unitary extension, all unused-coordinate Kraus branches, and arbitrary off-diagonal logical matrices. This is the coherent construction of Lemma 2.2, including the actual target embedding.

It also derives the Rains order constraints and collision upper bound from actual channels. A separately scoped extension derives the a/K bound and combines the three pointwise

bounds in (7). Source and configuration pins, fresh builds, empty-kernel replay, axiom reports, and proof-substitution controls are recorded separately for the two packages. Their original evidence and the previously fixed APPT audits retain their individual scopes.

The following distinctions are essential to the coverage statement.

Mathematical component	Verification status
Actual coherent subspace extraction, integer ranks and complete local channels	Lean
Actual channel-to-effect bounds, including the local-dimension cap	Lean, two scoped packages
Schur–Weyl identification, characters and Haar-average eigenvalues	Ordinary mathematical inputs and proofs
Support asymptotics, target-uniform minimax synthesis and limit laws	Ordinary mathematical proofs
Pure-input circuit synthesis and complexity estimates	Ordinary mathematical proof

In particular, the full paper is not represented as end-to-end Lean formalization. Finite exact calculations are additional diagnostics, not proofs of the unbounded representation-theoretic or probabilistic statements. The source archive gives theorem-to-source correspondence and literal hashes, so each verification claim can be checked against its actual endpoint.

6.3 Implementation and further questions

The pure-promise circuit of Proposition 5.5 is an effective implementation on the original register. The general mixed-state construction remains an existence theorem for a specified finite unitary and local channels. Efficient Schur transforms alone do not account for reversible sector indexing, variable Bell dimensions, and a resource-compatible completion [22, 23, 24]. An efficient implementation of that entire construction remains a separate problem.

The sharp minimax results fix D and s. Their finite inequalities remain valid when these parameters vary, but the inverse-moment asymptotic requires additional uniform estimates before such a joint limit can be claimed. The explicit conditions in (24) concern a different, information-spectrum route and do not replace those estimates. The exact reliability law at the zero-error boundary, cap-boundary Gaussian laws for arbitrary mixed inputs, and optimal circuit costs likewise remain open here.

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Predecessors and reproducibility. This manuscript consolidates the author's repository preprints on state-aware and state-independent extraction [26, 27]. Their original versions

are preserved. The finite-block construction at all blocklengths, target-uniform rank-budget minimax theorem, support/average two-moment comparison, and exact pure-promise resource accounting are identified separately in the contribution and source maps. The reproduction bundle contains this manuscript, its sources, the mathematical verification records, and the separately scoped Lean packages; it does not turn earlier preprints into distinct additional discoveries.

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